| Question | | | Answer | Marks | AO | Guidance |
|---|---|---|---|---|---|---|
| 7 | (a) | | When the boat is modelled as a particle, the size and shape of the boat are not taken into account in the model<br>Any rotation of the boat is neglected | **B1**<br><br>**[1]** | **3.3** | A sensible comment |
| 7 | (b) | | Resistance is $-300\mathbf{i}$ N<br>Newton's second law<br>$\left(\left(450\mathbf{i}+20\mathbf{j}\right)+\left(420\mathbf{i}-20\mathbf{j}\right)\right)-300\mathbf{i}=9000\mathbf{a}$<br><br>$\left[570\mathbf{i}=9000\mathbf{a}\right]$ | **M1**<br><br>**A1**<br><br>**[2]** | **1.1a**<br><br>**1.1b** | Sum of at least two forces seen in a N2L equation.<br>Also allow for scalar equation in the **i** direction only. Allow missing or incorrect resistance<br>Accept equivalent scalar equation and statement that there is no [resultant] force or no acceleration in the **j** direction) |
| 7 | (c) | | $\mathbf{a}=\frac{570}{9000}\mathbf{i}=0.0633\mathbf{i}\ \text{m s}^{-2}$ | **B1**<br><br>**[1]** | **2.5** | Must be vector. FT their equation(s) of motion |